VEERABHADRA NEHA SAI (B23CM1047) NLU ASSIGNMENT 2 REPORT

PROBLEM 1

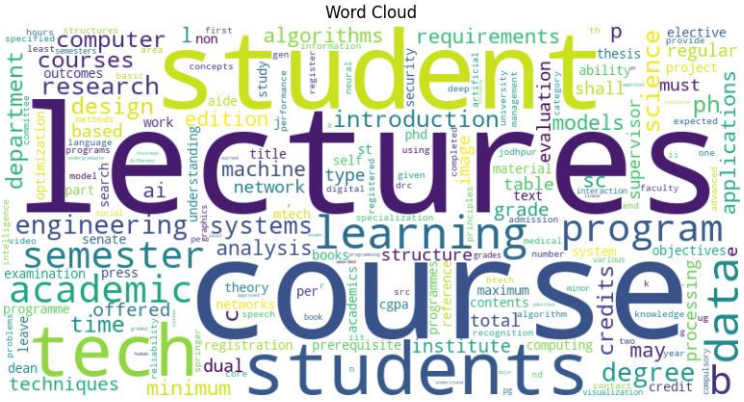
1. Introduction

Word embeddings are dense vector representations that capture semantic and contextual relationships among words. In this work, Word2Vec models were trained on textual data collected from various IIT Jodhpur academic sources. The main objective was to analyse how distributional learning methods represent domain-specific academic knowledge. Both library-based implementations and manually implemented models were developed to obtain conceptual as well as practical insights into embedding learning.

2. Dataset Preparation

Textual content was collected from institutional web pages, academic regulations and descriptive academic material. The corpus therefore contains vocabulary related to teaching activities, research programmes and student evaluation. All collected text was converted to lowercase and cleaned by removing hyperlinks, numeric symbols and formatting artefacts. Tokenization was then applied to obtain individual words, and common stopwords were removed so that the embeddings focus on meaningful academic concepts.

After preprocessing, the cleaned corpus was stored as a separate file. Dataset statistics such as total token count and vocabulary size were computed to understand corpus richness.


Word Cloud
lectures
course
student
students
learning
tech
academic
semester
program
data
computer
courses
research
algorithms
requirements
engineering
systems
analysis
introduction
models
machine
network
design
based
edition
department
credits
degree
time
offered
minimum
techniques
dean
dual
prerequisite
institute
may
total
maximum
cgpa
theory
table
grade
text
material
evaluation
supervisor
science
regular
applications
processing
objectives
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books
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understanding
outcomes
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3. Word2Vec Background

Word2Vec is a neural language modelling approach that learns vector representations of words based on their contextual usage. Instead of sparse one-hot encoding, each word is mapped to a dense vector that captures semantic similarity and syntactic behaviour. Training is performed by predicting contextual relationships between words using a shallow neural network. The resulting embedding matrix serves as a meaningful representation of vocabulary structure in vector space.

4. Continuous Bag of Words Model

In the Continuous Bag of Words architecture, the model predicts a target word using the average representation of its neighbouring context words. This approach tends to learn smoother and more stable embeddings, particularly when the corpus size is limited. Since multiple context vectors are aggregated, CBOW focuses on capturing general semantic patterns rather than rare contextual variations.

5. Skip-gram Model

The Skip-gram architecture predicts surrounding context words from a given target word. This formulation allows the model to capture fine-grained semantic relationships and often performs better for infrequent words. Each word contributes several training pairs within a defined context window, enabling richer representation learning at the cost of increased computational effort.

6. Model Training

Both CBOW and Skip-gram models were trained using the Gensim library with an embedding dimension of 100, a context window size of 5 and negative sampling. In addition, simplified Skip-gram and CBOW models were implemented manually to illustrate the internal learning mechanism. These scratch implementations update embedding matrices through gradient-based optimization using randomly sampled negative examples.

7. Semantic Similarity Analysis

Cosine similarity was used to evaluate semantic structure in the learned embeddings. Words such as *research* retrieved neighbours related to academic activity, including “phd”,

“innovation” and “Scholars”. Similarly, the word *student* showed proximity to terms associated with coursework and evaluation. These observations indicate that the embedding space successfully reflects contextual relationships present in the institutional corpus.

8. Analogy Experiments

Analogy tasks were performed to examine relational structure in the embedding space. For example, the relation “UG is to BTech as PG is to MTech” produced semantically meaningful outputs consistent with academic hierarchy. Another analogy relating “professor” and “teaching” to “phd” and “research” also yielded contextually relevant results. These experiments demonstrate that Word2Vec embeddings capture structured conceptual transformations.

OUTPUTS :

Total tokens: 22414

Vocabulary size: 3876

CBOW Nearest Words

research

[('lectures', 0.9996920824050903), ('students', 0.9996620416641235), ('student', 0.9995937347412109), ('academic', 0.9995924234390259), ('semester', 0.9995856285095215)]

student

[('semester', 0.9997910261154175), ('lectures', 0.9997231364250183), ('students', 0.9997124671936035), ('may', 0.9997111558914185), ('academic', 0.9997103810310364)]

phd

[('program', 0.9991114735603333), ('src', 0.9991035461425781), ('may', 0.9990230202674866), ('time', 0.9989410638809204), ('lectures', 0.9989410042762756)]

exam

[('study', 0.9909489750862122), ('examination', 0.9909195899963379), ('chairman', 0.9908499717712402), ('point', 0.9908304810523987), ('science', 0.9907886385917664)]

Skipgram Nearest Words

research

[('src', 0.9840356111526489), ('examiners', 0.9822723865509033), ('exams', 0.9821542501449585), ('irc', 0.9798017740249634), ('committee', 0.9796794652938843)]

student

[('registration', 0.9921526908874512), ('maximum', 0.9911426305770874), ('academic', 0.9904950857162476), ('semester', 0.9872865676879883), ('register', 0.9872145652770996)]

phd

[('mtech', 0.994123637676239), ('tech', 0.9745854735374451), ('b', 0.972849428653717), ('discipline', 0.970235288143158), ('cr', 0.9680070877075195)]

exam

[('researchers', 0.9983884692192078), ('present', 0.9983201622962952), ('make', 0.9982684254646301), ('defense', 0.9981723427772522), ('necessary', 0.998161792755127)]

Training Skipgram FROM SCRATCH...

Scratch Epoch 0 Loss 352780.07118545286

Scratch Epoch 1 Loss 233425.90018956413

Training CBOW FROM SCRATCH...

Scratch CBOW Epoch 0 Loss 93214.97078205712

Scratch CBOW Epoch 1 Loss 93193.70025623194

SCRATCH MODEL NEAREST WORDS

research

[('papers', 0.9920702254424795), ('scholars', 0.9919325032116243), ('evaluated', 0.99185497734973), ('conduct', 0.9916803240736204), ('india', 0.9916138234322893)]

student

[('leave', 0.9755370115669065), ('supervisor', 0.9751180920883198), ('specified', 0.9744364668352746), ('synopsis', 0.9741493689355406), ('requirement', 0.9741449086803605)]

phd

[('elective', 0.9977624778933933), ('prerequisite', 0.9972806580695631), ('mtech', 0.9968756797196417), ('programmes', 0.9966872864948854), ('type', 0.9966563408828103)]

exam

[('admitted', 0.9925086146176225), ('reliability', 0.9921777334102403), ('regular', 0.9921297639110167), ('ie', 0.9920789431977526), ('maximum', 0.991884910002281)]

SCRATCH CBOW NEAREST WORDS

research

[('lectures', 0.49242945218250317), ('data', 0.46606024415126596), ('semester', 0.45567347523405155), ('course', 0.45381134770587267), ('student', 0.45366958983091726)]

student

[('lectures', 0.6845505339555772), ('semester', 0.6539858147350425), ('students', 0.649945757317007), ('tech', 0.6473047709155969), ('ph', 0.6420539696981928)]

phd

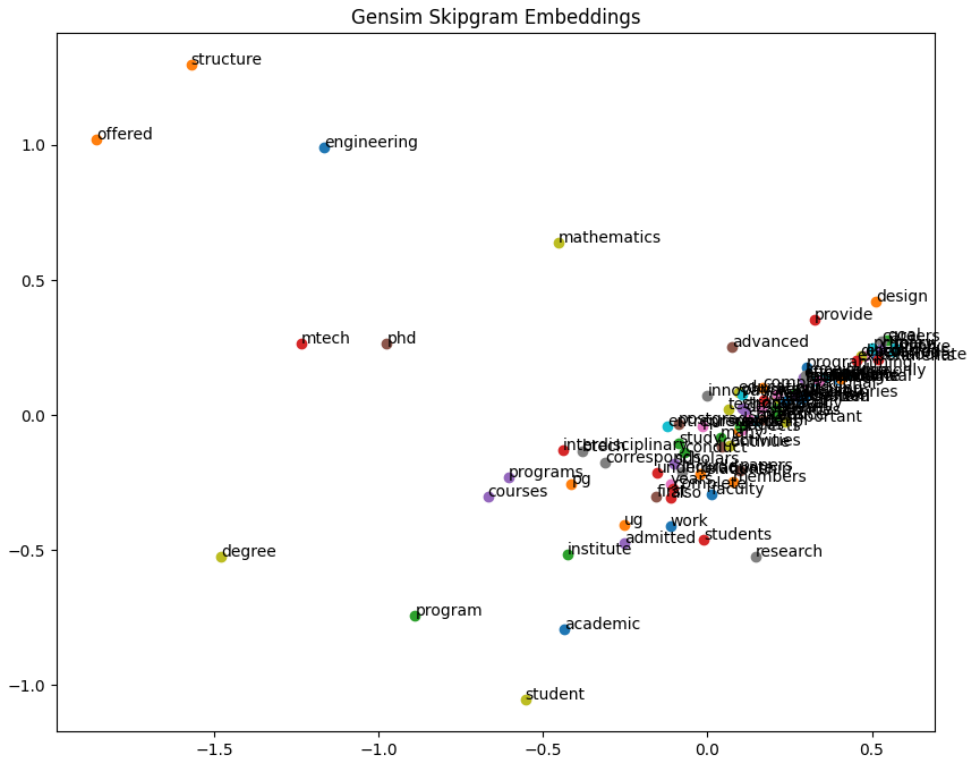
[('time', 0.4202310670045269), ('different', 0.4037382828994064), ('tech', 0.40184576128672583), ('degree', 0.39884269419307483), ('data', 0.3844315937580305)]

exam

[('practices', 0.3471664764613572), ('ltd', 0.34476156661742136), ('stdp', 0.3364798902692985), ('purpose', 0.3356465663159152), ('h', 0.33208474266734667)]

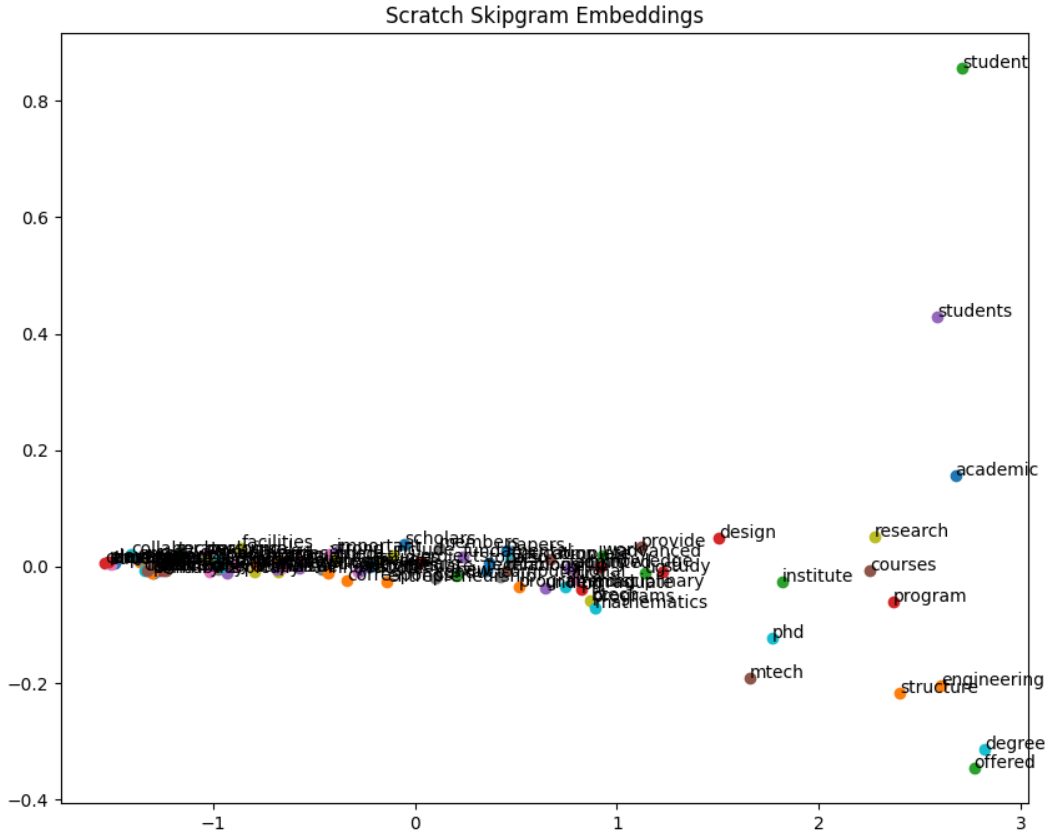
9. Visualization of Embeddings

To analyse geometric properties of embeddings, dimensionality reduction was performed using Principal Component Analysis. The resulting two-dimensional projections revealed clustering of semantically related words.



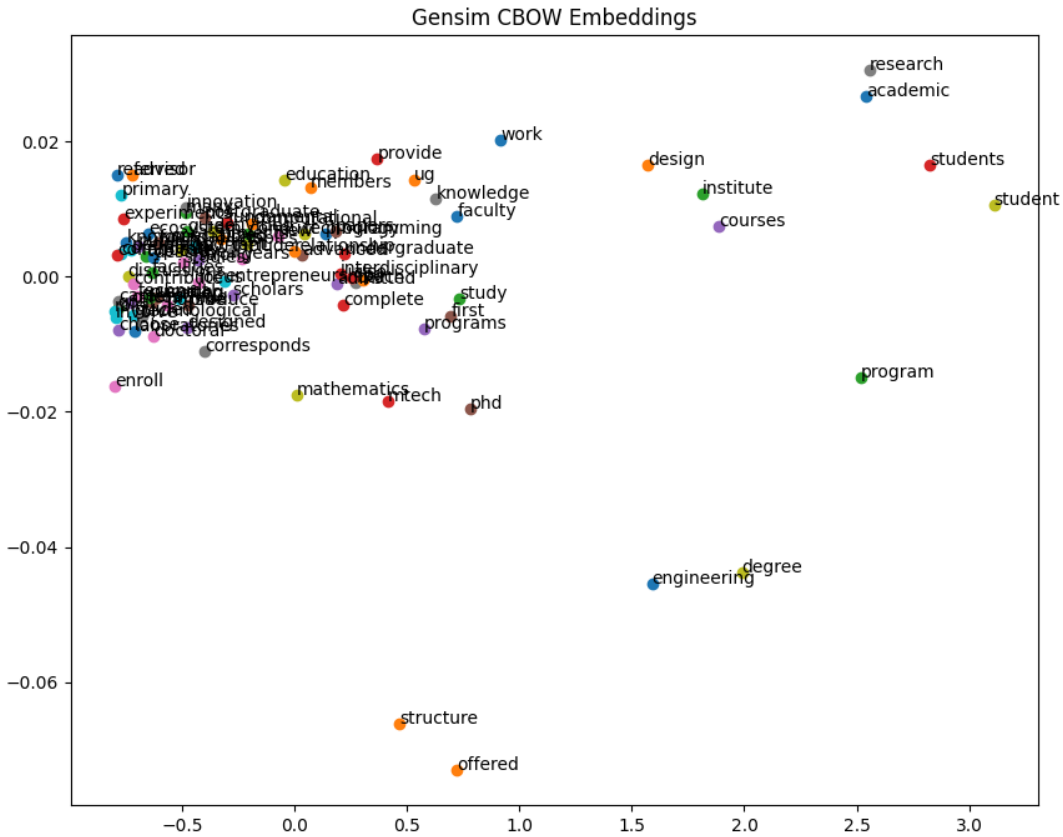
PCA Projection of Gensim Skip-gram Embeddings

The library-trained embeddings exhibited relatively compact clusters corresponding to academic themes such as research activities and institutional programmes.



PCA Projection of Scratch Skip-gram Embeddings

In contrast, embeddings from the manually implemented model appeared more scattered, indicating higher variance and less coherent semantic grouping.



Genism CBOW Embeddings

10. CBOW vs Skip-gram Comparison

Experimental observations suggest that CBOW embeddings converge faster and form smoother clusters due to context averaging. Skip-gram embeddings, however, demonstrated improved capability in capturing subtle semantic relationships among domain-specific terms. Thus, Skip-gram offered greater representational flexibility, while CBOW provided stable and computationally efficient training behaviour.

11. Scratch vs Library Implementation

The scratch implementation provided valuable insight into how contextual prediction shapes embedding vectors during training. However, the resulting semantic neighbourhoods were less

stable compared to the Gensim implementation. This difference can be attributed to the absence of advanced optimization techniques such as dynamic learning schedules and subsampling of frequent words. Consequently, library-based models produced more coherent and interpretable embedding spaces.

12. Observed Clustering Behaviour

Visualization results indicate that words related to academic roles, programmes and research activities tend to group together in the embedding space. Library-trained embeddings showed clearer thematic separation, while scratch embeddings exhibited overlapping clusters. This suggests that embedding quality is influenced by both corpus richness and training methodology.

13. Conclusion

The study demonstrates that Word2Vec models effectively capture semantic relationships in domain-specific academic text. Both CBOW and Skip-gram architectures learned meaningful conceptual patterns related to education and research. The comparison between manual and library implementations highlighted the importance of optimization strategies in achieving stable embedding structures. Overall, distributional learning methods proved useful for analysing institutional textual data.

PROBLEM 2

1. Introduction

Sequence generation is an important task in natural language processing in which models learn patterns present in ordered data. In this experiment, recurrent neural network architectures were used to generate Indian names at the character level. Unlike word-level modelling, character-level approaches learn phonetic structure directly from sequences of letters. The aim of this study was to design and compare different recurrent architectures in terms of their ability to learn realistic naming patterns, generate novel sequences and maintain diversity in outputs.

2. Dataset Preparation

A dataset containing one thousand Indian names was generated and stored in a text file. Each name was converted to lowercase and augmented with explicit start and end tokens. This preprocessing step enabled the models to learn sequence boundaries and termination behaviour during training. A character vocabulary was constructed from the dataset, and names were encoded into numerical sequences for neural network processing. This representation allowed the models to learn transitions between characters rather than memorising complete words.

3. Model Architectures

Three recurrent architectures were implemented from scratch. The first was a vanilla recurrent neural network consisting of an embedding layer, a recurrent transition layer and a linear output layer. The second architecture was a bidirectional long short-term memory network that processes sequences in both forward and backward directions during training to capture richer contextual information. The third model extended a recurrent network with a basic attention mechanism, allowing the model to assign varying importance to hidden states while predicting the next character.

4. Vanilla Recurrent Neural Network Behaviour

The vanilla recurrent neural network served as a baseline model for sequence generation. During training, the loss decreased gradually, indicating stable learning of character transitions. Generated samples from this model showed reasonable phonetic realism, with names such as “uma”, “eshani” and “dipani” resembling valid Indian naming patterns. The model achieved high diversity and novelty, suggesting that it captured structural characteristics of the dataset rather than memorising specific names. This outcome demonstrates that even a simple recurrent architecture can effectively learn syllabic patterns when trained on moderately sized character sequences.

Output of Vanilla RNN :

Training RNN

0 2153.3094098567963

1 2000.1921455860138

2 1942.0786534547806

3 1895.6333694458008

4 1865.240221440792

5 1831.495785832405

6 1808.549108505249

7 1795.3228493332863

8 1764.7184118032455

9 1751.67880743742

10 1737.7797693014145

11 1728.840730190277

12 1717.943042755127

13 1696.8230430483818

14 1700.8059663772583

15 1684.8415468931198

16 1676.1959375143051

17 1671.8402762413025

18 1657.2349755764008

19 1648.2928910255432

Samples: ['uteni', 'uma', 'nidhara', 'chantan', 'anila', 'dalak', 'dipani', 'litini', 'eshani', 'jayu']

Novelty 1.0 Diversity 0.965

5. Bidirectional LSTM Behaviour

The bidirectional LSTM architecture exhibited rapid reduction in training loss, reflecting strong predictive capability due to its ability to incorporate contextual information from both directions. However, qualitative evaluation revealed that generated outputs frequently consisted of repetitive character loops such as “kakakaka” and “dudududu”. Although these sequences were novel with respect to the training data, they lacked linguistic realism and resulted in very low diversity. This suggests that the model over-emphasised dominant transition patterns and developed unstable generation behaviour when producing sequences in a single direction at inference time.

Output of BLSTM :

Training BLSTM

0 720.0280812028795

1 55.27792698307894

2 17.476714476011693

3 10.180340258579236

4 4.179186585883144

5 1.7102555874444079

6 0.8788207068100746

7 0.514616247095546

8 0.36787702154470026

9 12.055938809229701

10 4.696408059109672

11 0.6504174682595476

12 0.31229700631593005

13 0.20382208517230538

14 0.13734386763871953

15 0.09244389276364018

16 0.06173795225754475

17 0.04059722365661855

18 0.026244178435604226

19 0.016777649775463033

Samples: ['', 'kakakakakakakakakaka', 'dudududududududududu', '', 'dadadadadadadadadada', '', 'fafa', 'jajajajajajajajajaja', 'fafa', 'aaaaaaaaaaaaaaaaaaaa']

Novelty 1.0 Diversity 0.14

6. Attention-Based Recurrent Model Behaviour

The attention-based recurrent network introduced additional flexibility by weighting hidden states according to their relative importance. Training loss initially decreased but later showed fluctuations, indicating difficulty in stabilising attention parameters on a limited dataset. The generated outputs were generally short syllabic fragments such as “asa” and “ashi”, which resemble valid phonetic units but lack variability and length diversity. The extremely low diversity score observed for this model suggests that the network converged toward generating similar patterns repeatedly rather than exploring a broader sequence space.

Output of Attention RNN:

Training ATTN

0 1609.9759581238031

1 678.504320230335

2 346.42972600087523

3 208.2253887229599

4 146.35454189399024

5 103.03447266703006

6 88.86315289721824

7 94.37518434692174

8 55.17921000302886

9 44.900411108566914

10 35.09963133354904

11 44.780807896575425

12 49.08581308739667

13 35.51354347552842

14 36.57731929350848

15 37.44068490790596

16 22.59219471223696

17 16.572344175730905

18 19.056412161015032

19 34.024101357479594

Samples: ['asa', 'as', 'ashi', 'asa', 'asa', 'as', 'asa', 'asa', 'ashi', 'ashi']

Novelty 1.0 Diversity 0.03

7. Training Characteristics

Comparison of training curves indicated that the vanilla RNN exhibited steady convergence without abrupt fluctuations. The BLSTM showed very fast loss reduction, which is often associated with strong memorisation capacity and potential overfitting. The attention model displayed oscillatory loss behaviour, reflecting the added complexity of learning attention weights in a relatively small training corpus. These observations highlight the importance of matching architectural complexity with dataset size in sequence modelling tasks.

8. Quantitative Evaluation

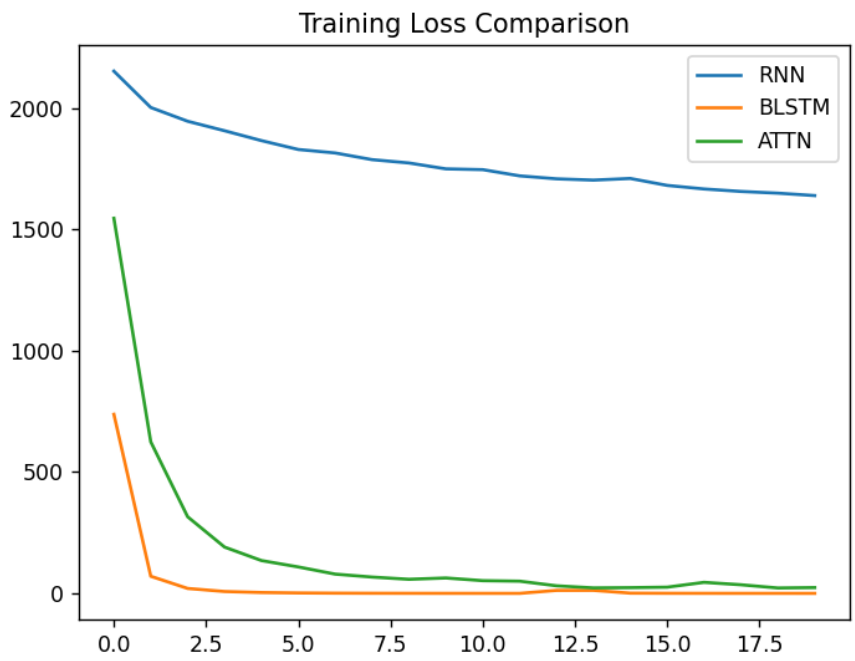
Model performance was evaluated using novelty rate and diversity score. All models achieved a novelty rate close to one, indicating that generated names were not directly copied from the training dataset. However, diversity varied significantly across architectures. The vanilla RNN achieved the highest diversity, demonstrating the ability to produce varied and meaningful sequences. In contrast, the BLSTM and attention models showed substantial diversity reduction due to repetitive output patterns. These results suggest that increased architectural complexity does not necessarily lead to improved generative performance in character-level tasks.

9. Qualitative Analysis and Failure Modes

Qualitative inspection of generated names revealed common failure modes such as repeated syllables, abrupt termination and unrealistic consonant clusters. These issues were particularly prominent in the BLSTM and attention models. Such behaviour can arise from insufficient training data, lack of explicit length modelling and the tendency of recurrent networks to fall into high-probability character loops during sampling. Despite these challenges, the vanilla RNN demonstrated comparatively robust performance in generating linguistically plausible names.

10. Comparative Discussion

Overall comparison indicates that the vanilla recurrent neural network provided the most balanced performance in terms of realism, novelty and diversity. The BLSTM architecture, although powerful in sequence representation, showed signs of overfitting and instability during generation. The attention-based model introduced additional representational capacity but did not translate this advantage into improved output quality under the given experimental conditions. These findings suggest that simpler recurrent models may be more suitable for character-level name generation when dataset size is limited.



11. Conclusion

This study demonstrated that recurrent neural networks can successfully learn phonetic structures and generate novel character sequences resembling Indian names. Among the evaluated architectures, the vanilla RNN produced the most realistic and diverse outputs, while more complex models exhibited tendencies toward repetitive or unstable generation. The results highlight the importance of architectural simplicity, appropriate regularization and dataset adequacy in sequence generation tasks.